

ASSESSMENT: C03-AT1: CASE STUDY

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0404

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Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

- 1. Find the mean, median, and mode.**
- 2. Find the range.**
- 3. Calculate the variance and standard deviation.**
- 4. Identify the outlier.**
- 5. Which measure of central tendency best represents the typical salary?
Explain why**

SOLUTION:

Given Data

The annual salaries (in ₹ lakhs) of 9 employees are:

4, 5, 5, 6, 7, 8, 8, 9, 30

Number of observations **N = 9**

The data is already arranged in ascending order.

Employee	Salary (₹ Lakhs)
1	4
2	5
3	5
4	6
5	7
6	8
7	8
8	9
9	30

Find the Mean:

Formula: Mean = $\frac{\sum X}{N}$

Where

- $\sum X$ = Sum of all observations
- N = Number of observations

Step 1: Find the Sum

$$4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30 = \mathbf{82}$$

Therefore,

$$\sum X = 82$$

Step 2: Apply the Formula

$$\begin{aligned}\text{Mean} &= \frac{82}{9} \\ &= \mathbf{9.11}\end{aligned}$$

Mean = 9.11 Lakhs

NOTE:

The mean is the arithmetic average of all salaries.

Although most employees earn between ₹4 lakh and ₹9 lakh, one employee earns ₹30 lakh.

B) Median

Formula

For odd number of observations,

$$\text{Median} = \left(\frac{N + 1}{2} \right)^{th} \text{ observation}$$

Since

$$N = 9$$

$$\frac{9 + 1}{2} = 5$$

The **5th observation** is:7

Median = 7 Lakhs

Note:

The median is the middle value of an ordered dataset.

Since there are **9 observations**, the fifth observation lies exactly in the center.

There are

- four salaries smaller than 7
- four salaries greater than 7

Therefore,

Median = ₹7 Lakhs

C) Mode

The mode is the value that occurs most frequently.

Let's count.

Salary	Frequency
4	1
5	2
6	1

Salary	Frequency
7	1
8	2
9	1
30	1

The highest frequency is **2**.

Two values have this frequency.

- 5
- 8

Mode = 5 Lakhs and 8 Lakhs

The data has **two modes**, so it is called **Bimodal Data**.

2: Find the Range:

The **range** is the simplest measure of **dispersion (spread)** in a dataset.

Formula

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Where:

- **Maximum Value** = Largest observation in the dataset
- **Minimum Value** = Smallest observation in the dataset

Step 1: Identify the Maximum Value

From the given data,

4, 5, 5, 6, 7, 8, 8, 9, 30

The **largest salary** is:30

Therefore,

Maximum Value = ₹30 Lakhs

Step 2: Identify the Minimum Value

The smallest salary :4

Therefore,

Minimum Value = ₹4 Lakhs

Step 3: Apply the Formula

$$\begin{aligned}\text{Range} &= 30 - 4 \\ &= 26\end{aligned}$$

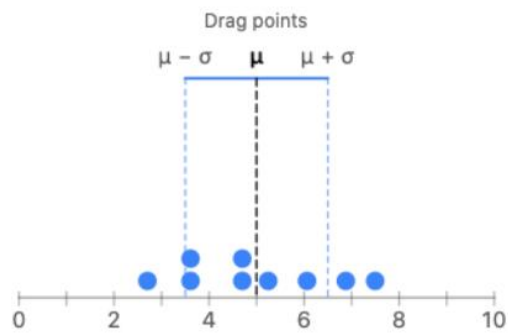
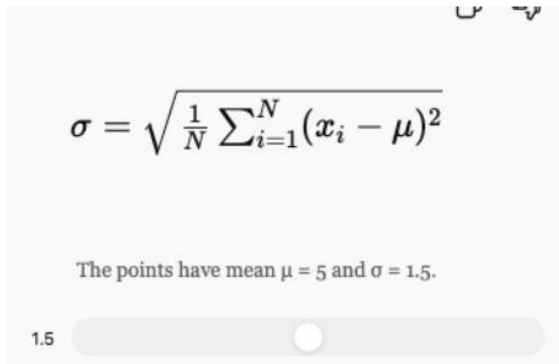
Answer: Range = 26 Lakhs

Interpretation

The salary distribution has a **range of ₹26 lakhs**, meaning the difference between the highest-paid employee and the lowest-paid employee is **₹26 lakhs**.

The lowest salary is **₹4 lakhs**, while the highest salary is **₹30 lakhs**. This large difference shows that salaries are **widely spread** within the company.

3. Calculate the Variance and Standard Deviation



Calculated Values

Mean

$$\mu = \frac{82}{9} = 9.11$$

Variance

$$\begin{aligned} \sigma^2 &= \frac{\sum (x - \mu)^2}{N} \\ &= \frac{512.87}{9} \\ &= 56.99 \end{aligned}$$

Standard Deviation

$$\begin{aligned} \sigma &= \sqrt{56.99} \\ &= 7.55 \end{aligned}$$

- **Variance = 56.99 Lakhs²**
- **Standard Deviation = 7.55 Lakhs**

The standard deviation of 7.55 lakhs indicates that the salaries are widely spread around the mean. This high value is mainly caused by the salary of ₹30 lakhs, which is much higher than the other salaries.

4: Identify the Outlier

What is an Outlier?

An outlier is an observation that is significantly different from the rest of the data. Outliers can affect statistical measures such as the mean, range, variance, and standard deviation.

Given Data

4, 5, 5, 6, 7, 8, 8, 9, 30

Most of the employee salaries lie between ₹4 lakhs and ₹9 lakhs. However, one employee earns ₹30 lakhs, which is much larger than all the other salaries.

Therefore,

Outlier = ₹30 Lakhs

5. Which Measure of Central Tendency Best Represents the Typical Salary? Explain Why

The median is the best measure of central tendency to represent the typical employee salary.

Reason

The given salaries are:

4, 5, 5, 6, 7, 8, 8, 9, 30

Most employees earn between ₹4 lakhs and ₹9 lakhs, whereas one employee earns ₹30 lakhs. The salary of ₹30 lakhs is an outlier because it is much higher than all the other salaries.

The mean is calculated using every value in the dataset. Therefore, the extremely high salary of ₹30 lakhs increases the mean from the actual salary level of most employees.

- Mean = ₹9.11 lakhs
- Median = ₹7 lakhs

The median represents the middle value of the ordered data and is not affected by the extreme salary. Hence, it gives a more accurate representation of what a typical employee earns.

Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

- 1. Calculate the mean and median.**
- 2. Find the range.**
- 3. Calculate the variance and standard deviation.**
- 4. Determine whether the data is likely to be right-skewed or left-skewed.**
- 5. Explain why the median may be more useful than the mean.**

Solution:

Given Data

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Number of observations:

$$N = 10$$

The data is already arranged in ascending order.

Request	Response Time (ms)
1	120
2	125
3	128
4	130
5	132
6	135
7	140
8	145
9	150

Request	Response Time (ms)
10	300

A) Mean

Definition

The mean is the arithmetic average of all observations. It is obtained by adding all the values and dividing the total by the number of observations.

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Where:

- $\sum X$ = Sum of all observations
- N = Number of observations

Step 1: Calculate the Sum

$$120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 \\ = 1505$$

Therefore,

$$\sum X = 1505$$

Step 2: Apply the Formula

$$\begin{aligned}\text{Mean} &= \frac{1505}{10} \\ &= 150.5\end{aligned}$$

Mean = 150.5 ms

The average response time of the website is 150.5 milliseconds.

B) Median

Definition

The median is the middle value of an ordered dataset. Since the number of observations is even (10), the median is the average of the 5th and 6th observations.

Formula

For even number of observations,

$$\text{Median} = \frac{\left(\frac{N}{2}\right)^{th} + \left(\frac{N}{2} + 1\right)^{th}}{2}$$

Step 1: Identify the Middle Values

5th observation = 132

6th observation = 135

Step 2: Calculate the Median

$$\text{Median} = \frac{132 + 135}{2}$$

$$\begin{aligned} &= \frac{267}{2} \\ &= 133.5 \end{aligned}$$

Median = 133.5 ms

2: Find the Range

Definition

The range is the difference between the highest and the lowest values in a dataset. It measures the overall spread of the data.

Formula

Range = Maximum Value – Minimum Value
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Step 1: Identify the Maximum Value

Maximum response time = 300 ms

Step 2: Identify the Minimum Value

Minimum response time = 120 ms

Step 3: Apply the Formula

$$\begin{aligned} \text{Range} &= 300 - 120 \\ &= 180 \end{aligned}$$

Range = 180 ms

Interpretation

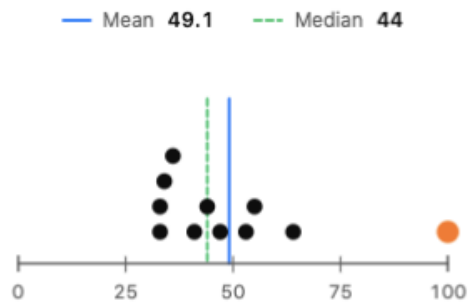
The response times vary over a span of 180 milliseconds, indicating a wide spread in the data. This large range is mainly due to the unusually high response time of 300 ms.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

- 1. Find Q1, Q2 and Q3.**
- 2. Calculate the IQR.**
- 3. Use the $1.5 \times \text{IQR}$ rule to identify any outliers.**
- 4. Find the mean and median.**
- 5. Explain how the outlier affects the mean.**

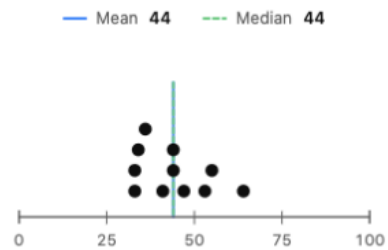


Outlier: mean moves 44 $\rightarrow$ 49.1; median stays 44

No outlier

High outlier

New data set



No outlier: mean is 44; median is 44

No outlier

High outlier

New data set

Given Data

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 55, 58, 60, 62, 65, 95

The data is already arranged in ascending order.

Student	Marks
1	42
2	45
3	48
4	50
5	52
6	52
7	55
8	58
9	60
10	62
11	65
12	95

Number of observation: $N=12$

Question 1. Find Q1, Q2 and Q3

What are Quartiles?

Quartiles divide an ordered dataset into four equal parts.

- Q1 (First Quartile): 25% of the data lies below this value.

- Q2 (Second Quartile): The median (50% of the data lies below this value).
- Q3 (Third Quartile): 75% of the data lies below this value.

Step 1: Find Q2 (Median)

Since there are 12 observations (even number),

Median = Average of the 6th and 7th observations.

6th observation = 52

7th observation = 55

$$\begin{aligned}Q_2 &= \frac{52 + 55}{2} \\&= \frac{107}{2} \\&= 53.5\end{aligned}$$

Q2 = 53.5

Step 2: Find Q1

The lower half of the data is

42, 45, 48, 50, 52, 52

Again, there are six observations.

Q1 = Average of the 3rd and 4th observations.

3rd observation = 48

4th observation = 50

$$Q_1 = \frac{48 + 50}{2}$$

$$= 49$$

$$Q_1 = 49$$

Step 3: Find Q3

Upper half:

55, 58, 60, 62, 65, 95

Middle observations:

$$3^{\text{rd}} = 60$$

$$4^{\text{th}} = 62$$

$$Q_3 = \frac{60 + 62}{2}$$

$$= 61$$

$$Q_3 = 61$$

2. Calculate the IQR

What is IQR?

The Interquartile Range (IQR) measures the spread of the middle 50% of the data.

It is less affected by extreme values than the range.

Formula

$$IQR = Q_3 - Q_1$$

Calculation

$$\begin{aligned} IQR &= 61 - 49 \\ &= 12 \end{aligned}$$

Answer: $IQR = 12$

Interpretation

The middle 50% of the students scored within a range of 12 marks.

3. Use the $1.5 \times IQR$ Rule to Identify Outliers

The $1.5 \times IQR$ Rule is commonly used to detect outliers.

Formula

Lower Limit

$$Q_1 - 1.5(IQR)$$

Upper Limit

$$Q_3 + 1.5(IQR)$$

Step 1

$$1.5 \times 12 = 18$$

Step 2

Lower Limit

$$49 - 18 = 3$$

Upper Limit: $61 + 18 = 79$

Step 3

Any value

- Less than 31
- Greater than 79

is an outlier.

The data contains

95

Since: $95 > 79$

95 is an Outlier

4. Find the Mean and Median

Mean

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Sum

$$\begin{aligned} &42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95 \\ &= 684 \end{aligned}$$

Calculation

$$\begin{aligned}\text{Mean} &= \frac{684}{12} \\ &= 57\end{aligned}$$

Mean = 57

Median

Already calculated

$$\text{Median} = 53.5$$

5. Explain How the Outlier Affects the Mean

The mark 95 is an outlier because it is much higher than the marks obtained by the other students.

Most students scored between 42 and 65, but one student scored 95. Since the mean uses every value in the dataset, the outlier increases the average.

- Mean = 57
- Median = 53.5

The mean becomes higher than the median because it is pulled upward by the high score of 95.

The median is the middle value and is not affected by the outlier. Therefore, it represents the typical student performance more accurately.

SOLUTION:

Given Data

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

The data is already arranged in ascending order.

Day	Number of Orders
1	18
2	20
3	22
4	22
5	24
6	25
7	26
8	28
9	30
10	35

Number of observations:

$$N = 10$$

Question 1. Calculate the Mean, Median, and Mode

A) Mean

Definition

The mean is the arithmetic average of all observations. It is calculated by dividing the sum of all observations by the total number of observations.

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Where:

- $\sum X$ = Sum of all observations
- N = Number of observations

Step 1: Calculate the Sum

$$18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35 = 250$$

Step 2: Apply the Formula

$$\begin{aligned}\text{Mean} &= \frac{250}{10} \\ &= 25\end{aligned}$$

Answer **Mean = 25**

Interpretation

The online store receives an average of 25 orders per day over the 10-day period.

B) Median

Definition

The median is the middle value of an ordered dataset.

Since the number of observations is even (10), the median is the average of the 5th and 6th observations.

Formula

$$\text{Median} = \frac{\left(\frac{N}{2}\right)^{th} + \left(\frac{N}{2} + 1\right)^{th}}{2}$$

Step 1: Identify the Middle Values

5th observation = 24

6th observation = 25

Step 2: Calculate

$$\begin{aligned}\text{Median} &= \frac{24 + 25}{2} \\ &= \frac{49}{2} \\ &= 24.5\end{aligned}$$

Answer **Median = 24.5**

Interpretation

The median indicates that half of the recorded days had 24.5 or fewer orders, while the other half had 24.5 or more orders.

C) Mode

Definition

The mode is the observation that occurs most frequently.

Orders	Frequency
18	1
20	1
22	2
24	1
25	1
26	1
28	1
30	1
35	1

The value 22 occurs twice, while all other values occur only once.

Answer **Mode = 22**

2. Find the Variance and Standard Deviation

Definition

Variance measures the average squared distance of each observation from the mean. Standard deviation is the square root of the variance and measures the average spread of the data.

Formula (Population Variance)

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

Where

- $\mu = 25$
- $N = 10$

Calculation Table

Orders (x)	$(x - \mu)$	$((x - \mu)^2)$
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1

Orders (x)	(x-\mu)	((x-\mu)^2)
28	3	9
30	5	25
35	10	100

Sum of Squared Deviations

$$49 + 25 + 9 + 9 + 1 + 0 + 1 + 9 + 25 + 100 = 228$$

Variance

$$\begin{aligned}\sigma^2 &= \frac{228}{10} \\ &= 22.8\end{aligned}$$

Answer **Variance = 22.8**

Standard Deviation

$$\begin{aligned}\sigma &= \sqrt{22.8} \\ &= 4.77\end{aligned}$$

Answer **Standard Deviation = 4.77**

3. Calculate the Coefficient of Variation (CV)

Definition

The Coefficient of Variation (CV) measures the variability of the data relative to its mean. It is expressed as a percentage and is useful for comparing the consistency of different datasets.

Formula

$$CV = \frac{\sigma}{\mu} \times 100$$

Where:

- σ = Standard Deviation
- μ = Mean

Substitute the Values

Mean: $\mu = 25$

Standard Deviation

$$\sigma = 4.77$$

$$CV = \frac{4.77}{25} \times 100$$
$$= 19.08\%$$

Answer: $CV = 19.08\%$

Interpretation

A coefficient of variation of 19.08% indicates that the daily orders have moderate variability.

Question 4. What Does the Standard Deviation Tell the Company?

Answer

The standard deviation (4.77 orders) indicates the average variation of daily orders from the mean value of 25 orders.

A standard deviation of 4.77 means that the number of daily orders generally differs from the average by about 5 orders.

For the company, this information is useful because:

- It helps estimate the expected daily variation in customer demand.
- It supports inventory planning by indicating how much stock may be needed.
- It assists in staff scheduling by predicting busy and less busy days.
- It improves delivery planning and warehouse management.

5. Would You Describe the Order Data as Highly or Moderately Variable?

The order data can be described as moderately variable.

- Explanation
- The mean number of daily orders is 25, while the standard deviation is 4.77. This means that the daily orders usually vary by approximately 5 orders from the average.
- Additionally, the Coefficient of Variation (19.08%) is relatively low. A CV below 20% generally indicates moderate variability, suggesting that the data is fairly consistent.
- Although there is a maximum value of 35 orders, it is not extremely far from the other observations and does not act as a significant outlier.

QUESTION:5 CASE STUDY

compare two data science models

two machine-learning models have the following prediction errors:

MODEL A:2,3,3,4,4,5,5,6

MODEL B:1,1,2,4,5,7,8,9

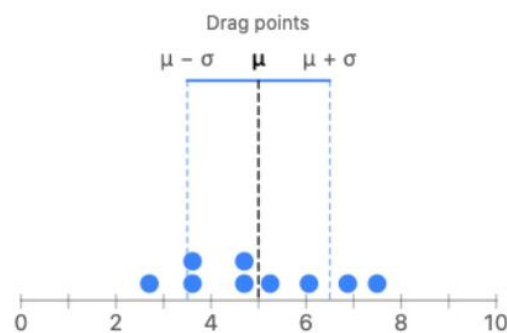
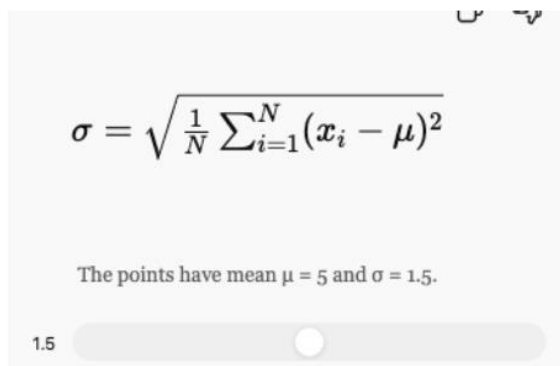
1.calculate the mean error for both models.

2.calculate the variance for both models.

3.calculate the standard deviation for both.

4.which model has more consistent predictions?

5.can the model with the lower mean error necessarily be considered better ? Explain.



SOLUTION:

Given Data

Two machine learning models have the following prediction errors:

Model A

2, 3, 3, 4, 4, 5, 5, 6

Model B

1, 1, 2, 4, 5, 7, 8, 9

Number of observations:

$$N = 8$$

1. Calculate the Mean Error for Both Models

What is Mean Error?

The mean error is the average prediction error made by a model. It is calculated by adding all the prediction errors and dividing by the total number of observations.

Model A

Step 1: Calculate the Sum

$$2+3+3+4+4+5+5+6=32$$

Step 2: Apply the Formula

$$\text{Mean} = 32/8$$

$$=4$$

$$\text{Mean Error of Model A} = 4$$

Model B

Step 1: Calculate the Sum

$$1+1+2+4+5+7+8+9=37$$

Step 2: Apply the Formula

$$\text{Mean} = 37/8$$

=4.625

Mean Error of Model B = 4.625

2. Calculate the Variance for Both Models

Definition

Variance measures how much the prediction errors are spread around the mean.

A smaller variance indicates that the prediction errors are more consistent.

Formula

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

Model A

Mean: $\mu = 4$

Error (x)	$(x - \mu)$	$((x - \mu)^2)$
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1

Error (x)	(x-\mu)	((x-\mu)^2)
5	1	1
6	2	4

Sum of squared deviations:

$$4 + 1 + 1 + 0 + 0 + 1 + 1 + 4 = 12$$

Variance

$$\sigma^2 = \frac{12}{8}$$

$$= 1.5$$

Variance of Model A = 1.5

Model B

Mean: $\mu = 4.625$

Error (x)	(x-\mu)	((x-\mu)^2)
1	-3.625	13.1406
1	-3.625	13.1406
2	-2.625	6.8906
4	-0.625	0.3906
5	0.375	0.1406

Error (x)	(x-\mu)	((x-\mu)^2)
7	2.375	5.6406
8	3.375	11.3906
9	4.375	19.1406

Sum of squared deviations

$$= 69.875$$

Variance

$$\sigma^2 = \frac{69.875}{8}$$

$$= 8.734$$

Variance of Model B = 8.734

3. Calculate the Standard Deviation for Both Models

Definition

The standard deviation measures the average spread of prediction errors around the mean.

Formula

$$\sigma = \sqrt{\sigma^2}$$

Model A

Variance:= 1.5

Standard Deviation

$$\begin{aligned}\sigma &= \sqrt{1.5} \\ &= 1.225\end{aligned}$$

Standard Deviation of Model A = 1.225

Model B

Variance:= 8.734

Standard Deviation

$$\begin{aligned}\sigma &= \sqrt{8.734} \\ &= 2.955\end{aligned}$$

Standard Deviation of Model B = 2.955

4. Which Model Has More Consistent Predictions?

Model A has more consistent predictions.

Explanation

Consistency refers to how closely the prediction errors are grouped around the mean. A model with a lower variance and lower standard deviation is considered more consistent because its errors vary less.

Comparison:

Measure	Model A	Model B
Mean Error	4.00	4.625
Variance	1.5	8.734
Standard Deviation	1.225	2.955

Model A has:

- Lower variance
- Lower standard deviation
- Prediction errors that are closer to each other

Therefore, Model A produces more stable and reliable predictions than Model B.

5. Can the Model with the Lower Mean Error Necessarily Be Considered Better? Explain.

Answer

No, a model with the lower mean error cannot always be considered better.

Explanation

Although Model A has a lower mean error (4.00) than Model B (4.625), the quality of a machine learning model should not be judged only by its mean error.

Several other factors should also be considered:

- Variance and Standard Deviation: Lower values indicate more consistent predictions.
- Prediction Stability: A model that gives similar errors across different observations is generally more reliable.
- Accuracy: The overall correctness of predictions.
- Precision and Recall: Important for classification problems where identifying positives correctly matters.
- F1-Score: Balances precision and recall.
- Application Requirements: Some applications may tolerate larger errors, while others require extremely accurate predictions.

In this case, Model A not only has a lower mean error but also a much lower variance and standard deviation, making it both more accurate and more consistent